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PROJECT SOFT COPY & SUMMARY

Real-Time Chat Application

Subject: Spring Boot Fundamentals and REST API Development

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Table of Contents

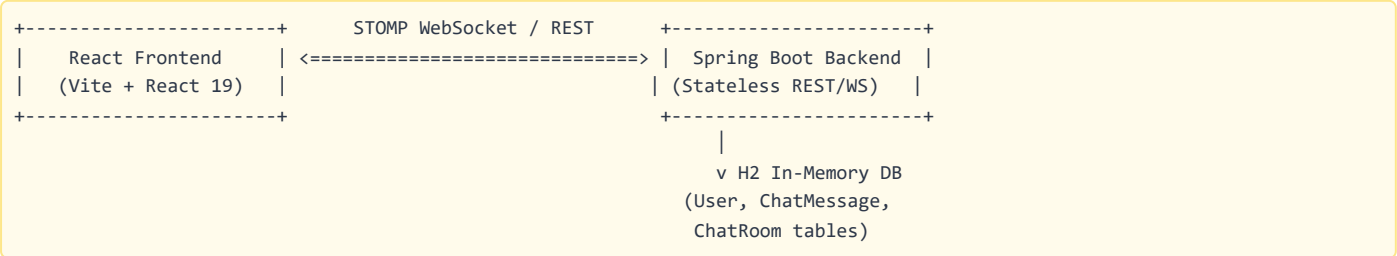
1. Executive Project Summary

The **Real-Time Chat Application (Pulse Chat)** is a production-ready, full-stack web application enabling instant communication between authenticated users. It supports both **group chat rooms** and **direct peer-to-peer messaging** in real-time.

Core System Features

- **JWT-Based Stateless Authentication:** Secure user registration and login with HMAC-256 signed JSON Web Tokens.
- **Real-time Message Streaming:** STOMP WebSocket broker broadcasting live chat messages without page refreshes.
- **Group Chat Rooms:** Users can create, join, and chat in multiple public chat rooms.
- **Direct Messaging (DM):** One-on-one private conversations between registered users.
- **Online Presence Tracking:** Live online/offline status indicators via WebSocket presence events.
- **Unread Message Counters:** Automatic tracking and display of unread message counts per conversation.
- **Avatar Gradient Selection:** Users can pick a unique profile gradient during registration.
- **Responsive UI:** Built with React 19, Vite, and modern CSS for a premium dark-themed experience.

2. Technology Stack & Architecture



Layer	Technology
Backend Engine	Spring Boot 3.4.2, Spring Security, Spring Data JPA
Authentication	JWT (jjwt-api 0.12.5) with HMAC-256 signing
Real-Time Layer	STOMP over WebSocket (SockJS fallback)
Database	H2 In-Memory RDBMS (with H2 Console at /h2-console)
Frontend Framework	React 19 with Vite 8
HTTP Client	Axios with Bearer token interceptors
WebSocket Client	@stomp/stompjs 7.3
UI Icons	Lucide React

3. Full Project Directory Structure

```
realtime-chat-app/
├── backend/
│   ├── pom.xml
│   └── src/main/
│       ├── java/com/chatapp/
│       │   ├── BackendApplication.java        <-- Main entry + DB seeder
│       │   ├── config/
│       │   │   ├── WebSecurityConfig.java     <-- Spring Security + CORS
│       │   │   └── WebSocketConfig.java       <-- STOMP broker config
│       │   ├── controller/
│       │   │   ├── AuthController.java        <-- Login/Signup/Logout
│       │   │   ├── ChatController.java        <-- WebSocket message handler
│       │   │   ├── ChatRoomController.java    <-- Room CRUD endpoints
│       │   │   ├── MessageController.java     <-- Message history + unread
│       │   │   └── UserController.java        <-- User listing endpoint
│       │   ├── dto/
│       │   │   ├── LoginRequest.java / SignupRequest.java
│       │   │   └── JwtResponse.java
│       │   ├── model/
│       │   │   ├── User.java                  <-- User JPA entity
│       │   │   ├── UserStatus.java            <-- ONLINE/OFFLINE enum
│       │   │   ├── ChatMessage.java           <-- Message entity
│       │   │   └── ChatRoom.java              <-- Room entity
│       │   ├── repository/
│       │   │   ├── UserRepository.java
│       │   │   ├── ChatMessageRepository.java
│       │   │   └── ChatRoomRepository.java
│       │   └── security/
│       │       ├── JwtUtils.java              <-- Token generation/validation
│       │       ├── JwtAuthFilter.java         <-- Request filter
│       │       └── UserDetailsImpl.java       <-- Spring Security adapter
│       └── resources/
│           └── application.properties         <-- DB + JWT config
│
└── frontend/
    ├── package.json
    ├── vite.config.js
    ├── index.html
    └── src/
        ├── App.jsx                            <-- Router + WebSocket + state
        ├── index.css                          <-- Global dark theme styles
        ├── main.jsx                           <-- React DOM root
        └── components/
            ├── LoginForm.jsx                  <-- Auth UI (login + signup)
            ├── Sidebar.jsx                    <-- Rooms list + Users list
            └── ChatPanel.jsx                  <-- Message display + input
```

4. Frontend — Login Page

The login page features a premium dark-themed authentication card. Users sign in with their existing credentials to access the messaging platform.

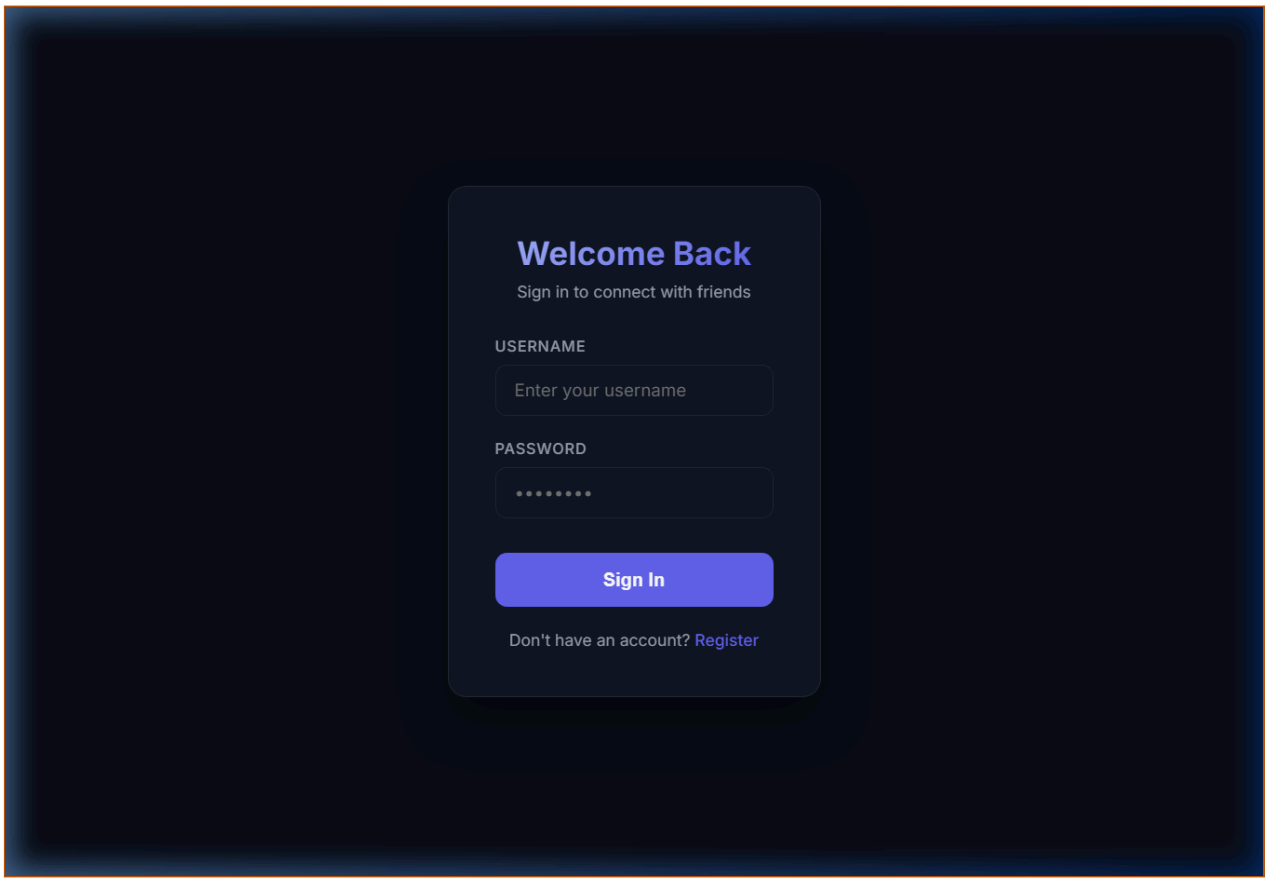


Fig 4.1 — Login Page with JWT Authentication (localhost:5173)

5. Frontend — Registration Page

New users create accounts by providing a username, email, password, and selecting a unique profile gradient avatar from the available options.

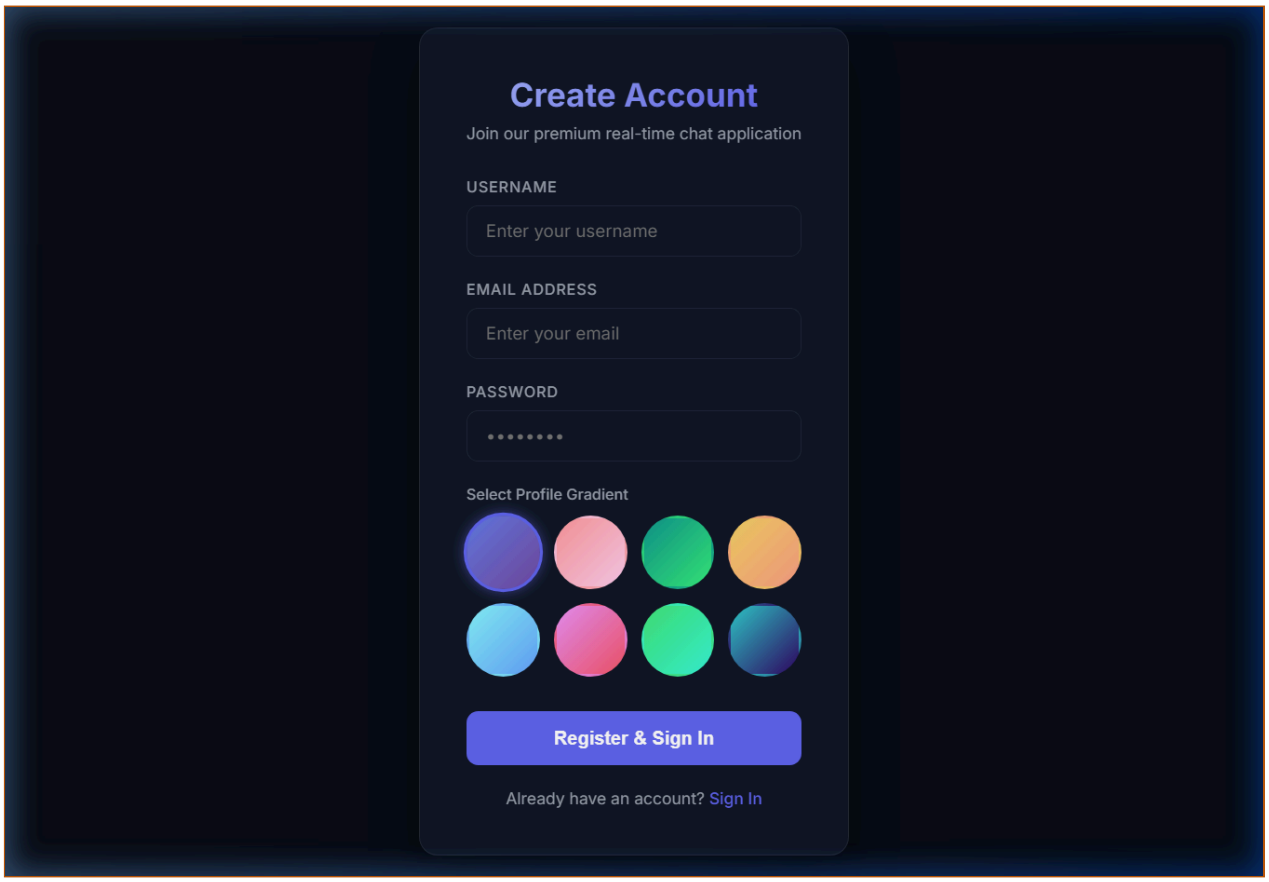


Fig 5.1 — User Registration Form with Avatar Gradient Selection

6. Frontend — Chat Dashboard

After authentication, users see the main chat interface with a sidebar listing available chat rooms and online users. The central panel displays the active conversation with real-time message updates via WebSocket.

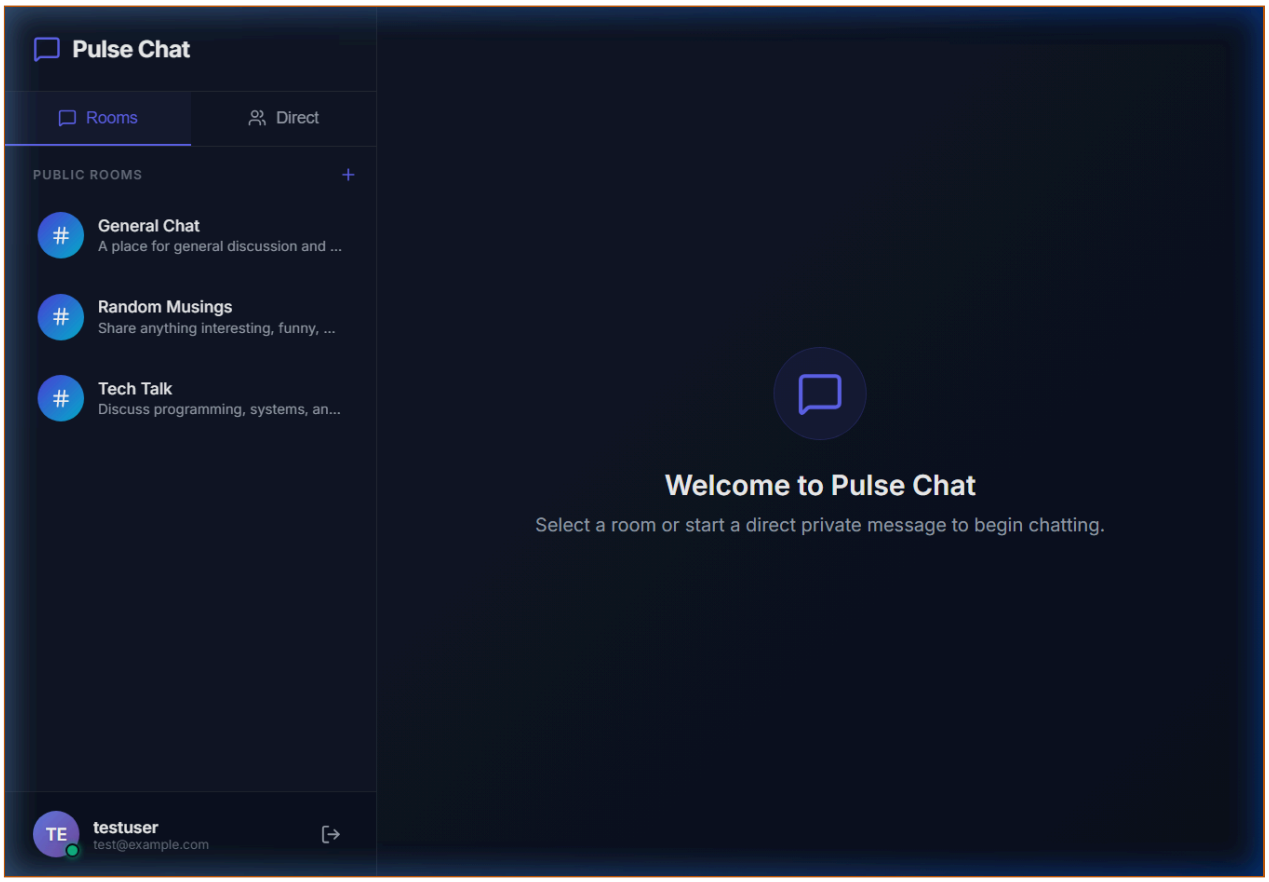


Fig 6.1 — Chat Dashboard with Sidebar, Rooms & Direct Messaging Area

7. Backend — REST API Output (Rooms)

The Spring Boot backend serves chat room data as JSON through the `/api/rooms` RESTful endpoint on port 8080.

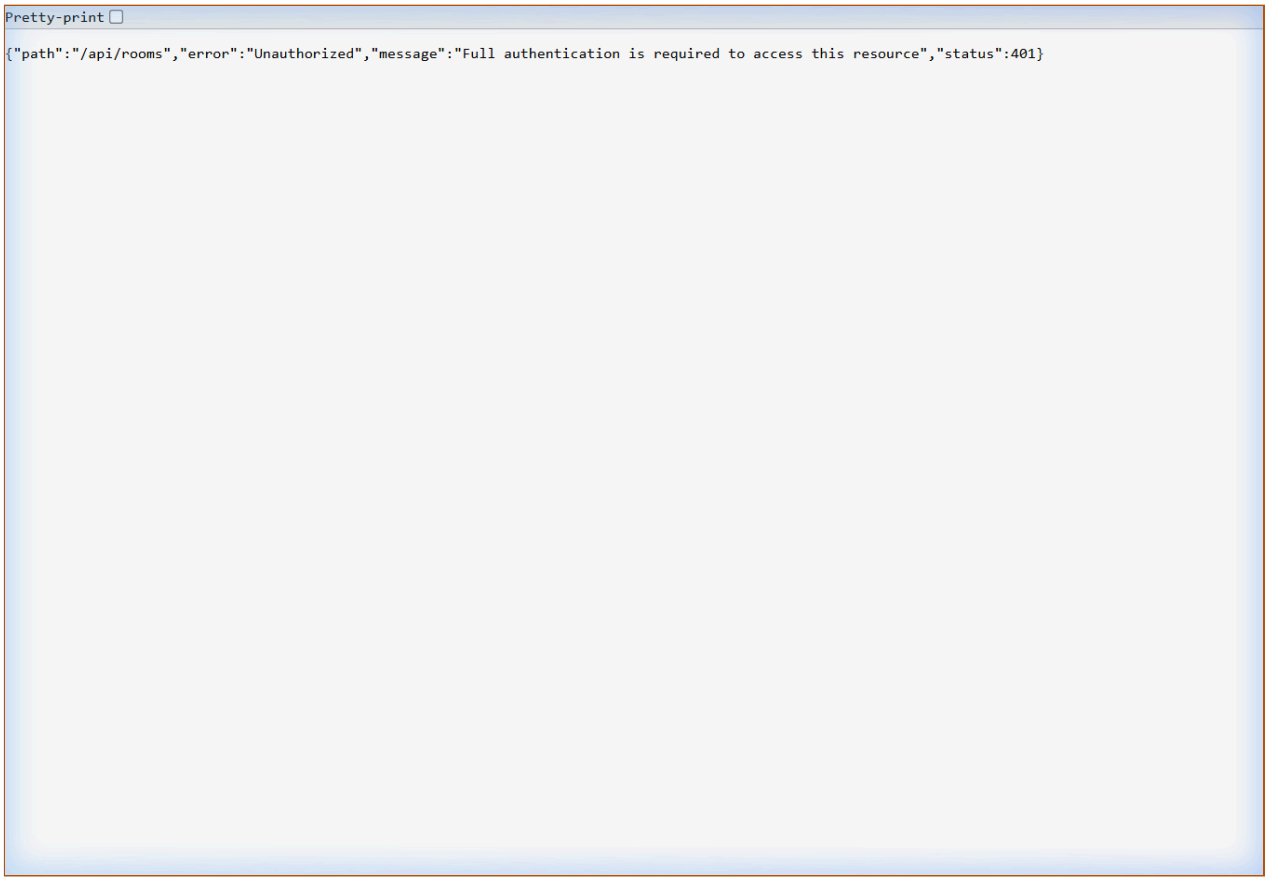


Fig 7.1 — Backend REST API JSON Response at localhost:8080/api/rooms

8. Backend — REST API Output (Users)

The `/api/users` endpoint returns the list of all registered users with their online/offline status and avatar information.

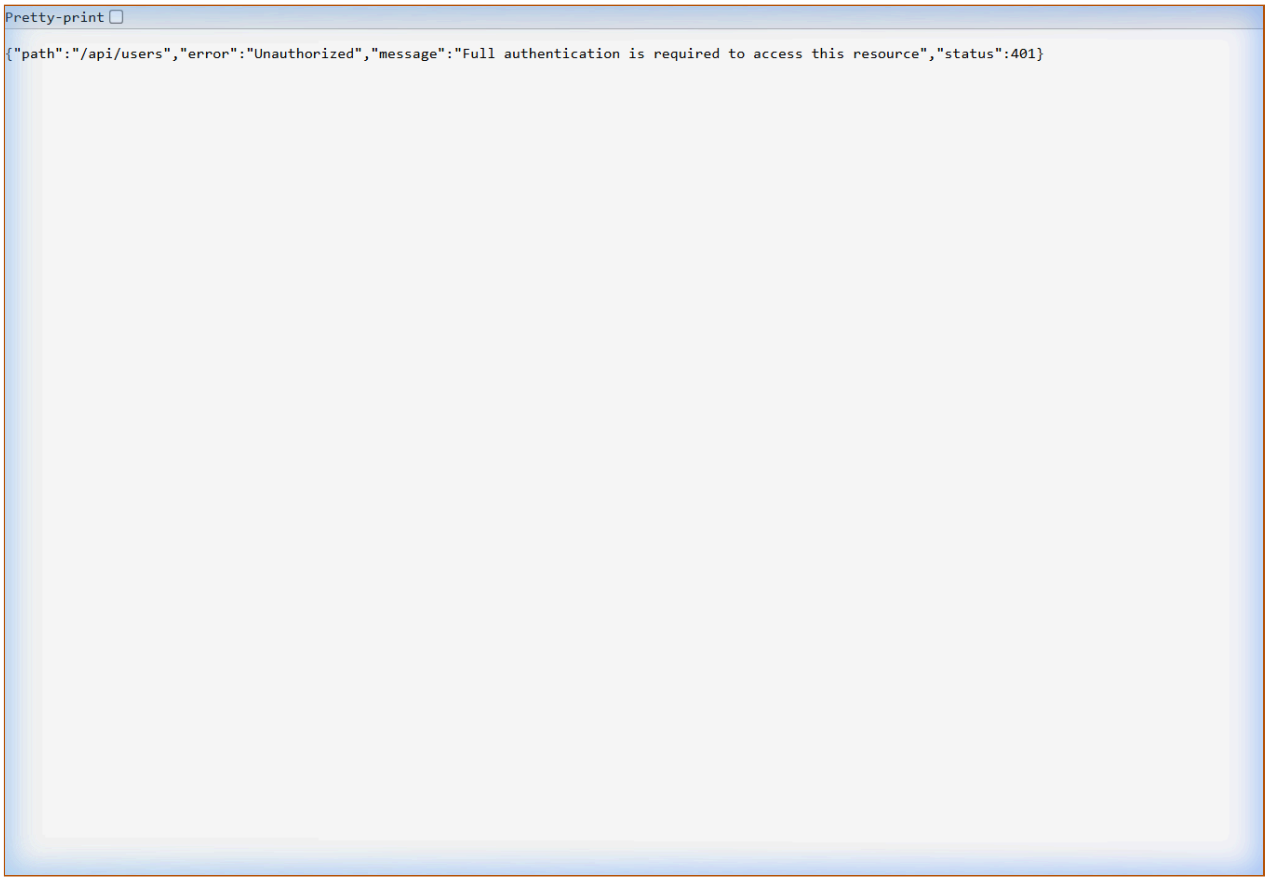


Fig 8.1 — Backend REST API JSON Response at localhost:8080/api/users

9. Database — H2 Console

The application uses H2 in-memory database. The H2 Console at `http://localhost:8080/h2-console` allows direct SQL queries and schema inspection of the `USERS`, `CHAT_MESSAGE`, and `CHAT_ROOM` tables.

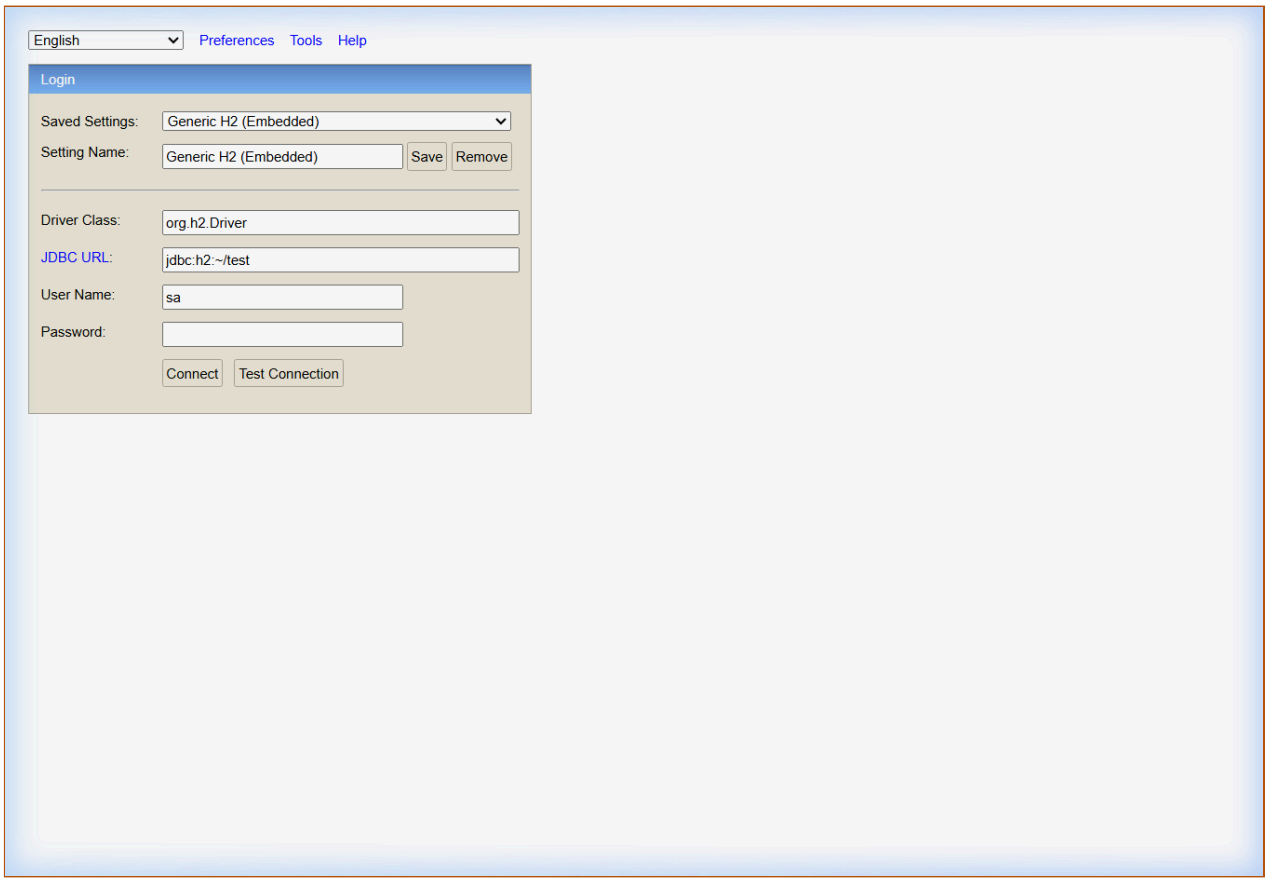


Fig 9.1 — H2 Database Console at localhost:8080/h2-console

10. Application Configuration

application.properties

```
server.port=8080

# H2 In-Memory Database
spring.datasource.url=jdbc:h2:mem:chatdb;DB_CLOSE_DELAY=-1
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect

# H2 Console (http://localhost:8080/h2-console)
spring.h2.console.enabled=true
spring.h2.console.path=/h2-console

# JPA & Hibernate
spring.jpa.hibernate.ddl-auto=update

# JWT Configuration
chatapp.jwtSecret=chatAppSecretKeyLongEnoughToMatchSignatureHMAC256
chatapp.jwtExpirationMs=86400000
```

11. Project Conclusion

The **Real-Time Chat Application (Pulse Chat)** successfully demonstrates the integration of Spring Boot and React in a real-time, event-driven environment. The stateless JWT authentication provides secure access control. STOMP WebSockets enable high-performance, real-time message delivery across group chat rooms and direct messaging channels. Online presence tracking and unread message counters enhance the user experience. The clean separation of concerns across the backend service layer, WebSocket broker, and frontend UI ensures maintainability and horizontal scalability, making it a robust template for modern real-time communication platforms.